Michele Valsecchi – Curriculum Vitae

Nationality: Italian – **Date of birth:** 03/16/1996 – **E-mail:** valsecchi.michele@outlook.com
LinkedIn – Google Scholar – ORCiD – GitHub

PROFESSIONAL SUMMARY

I am a materials scientist/engineer specialized in **complex fluids and soft matter.** I combine **theory, computational physics and experiments** to develop physical insight and help design novel advanced materials. I am passionate about **collaborative**, **interdisciplinary and international science**.

RESEARCH

Columbia University - Postdoctoral Research Scientist **2023 – Present**

I have **co-supervised, trained and mentored** 8 PhD and 10+ BSc/MSc students. I **helped write 2 NSF grant proposals** (1 funded, $500k, #2609474, PI S. Kumar; 1 pending, PI S. Kumar & J. Fish).

Funding:

- **Department of Energy – Genesis Mission**: "AI-Enabled Physical Operating System for Bio-programmable Matter" (2026 – 2027) – PI Prof. Kyle Bishop
- **National Science Foundation #2225290**: "Multiscale Data-Physics Models for the Critical Role of Interfaces in Overmolded Thermoplastic Parts" (2023 – 2026) – PI Prof. Sanat Kumar

Active research areas:

- Multi-physics simulation of plastics processing, DNA self-assembly and gel thermodynamics
- Polymer degradation, with a focus on nanoplastics release
- Statistical mechanics of polymers and complex fluids in and out of equilibrium
- Closed-loop AI-driven experimentation

TECHNICAL SKILLS

Theory:
- Statistical mechanics
- Polymer physics
- Self-consistent field theory

Computational physics:
- Molecular Dynamics
- Monte Carlo
- Density Functional Theory (PySCF)

Programming:
- Python
- HPC/SLURM
- Unix/Bash
- Git

Experimental:
- SAXS/WAXS
- DSC/TGA
- GC-MS
- Raman

EDUCATION

Imperial College London **2019 – 2023**
PhD in Theory and Simulation of Materials
Fully funded by Procter & Gamble. Supervised by George Jackson and Amparo Galindo. **Thesis**: "Molecular modelling and statistical thermodynamics of semi-crystalline polymers: network constraints, fluid sorption and surface melting". **Wrote 1 EPSRC grant** to optimize polymer liners for H_2 storage (PIs MG de Angelis, G. Jackson, A. Galindo - not funded).

Imperial College London **2018 – 2019**
MSc in Theory and Simulation of Materials
Grade: Distinction. **Thesis**: "Modelling the thermodynamic properties of aqueous PEG mixtures". Focus on computational physics (MD, MC, Free Energy Methods, DFT, AI/ML) and **grant writing**.

Stanford University **Summer 2017**
Undergraduate Summer Visitor – Organic Chemistry Intensive
Grade: A+ – Top 1%. Experience with polymer synthesis and analytical techniques

Politecnico di Milano **2015 – 2018**
BSc in Materials and Nanotechnology Engineering
Grade: 110/110 cum Laude. **Thesis**: "Anodic intercalation of graphite: application to the production of graphene and graphene oxide-based systems"

TEACHING

I am an **Associate Fellow** of **Advance HE** – the UK Higher Education Academy – since 2023.

Columbia University – Lecturer **2024 – 2025**
Co-designed and delivered the Polymer Physics graduate course in the Department of Chemical Engineering. Prepared **original lecture material** and examined about 20 students.

Imperial College London – Graduate Teaching Assistant **2018 – 2023**
400+ h **marking, tutoring and mentoring** for courses in the Materials and Chemical Engineering Departments. Honored by the **"Outstanding GTA" Student Choice Award**.

SELECTED PUBLICATIONS (17 published/in press, 7 first-author, 184 total citations, h-index 7)

[1] Valsecchi M, Ninarello A, Zaccarelli E, Kumar SK. "Phase transitions of polymer networks under tension". In press in *Macromolecules,* (**2026**)

[2] Valsecchi M, Fall WS, Meyer H, Grest GS, Kumar SK. "Molecular insights on the thermal welding of semicrystalline polymers". *Physical Review Letters* (**2026**), 137, 098101 – **On the Cover**

[3] Valsecchi M, Kalghatgi SR, Mendez NF, Lincoln C, Rorrer NA, Kumar SK. "Role of amorphous chains in nanoplastics formation from semicrystalline polymers". *ACS Macro Letters* (**2026**), 15(9), 1228-1232

[4] Mendez NF, Sharma V, Valsecchi M, Pai V, Lee JK, Schadler LS, Müller AJ, Watson-Sanders S, Dadmun M, Kumaraswamy G, Kumar SK. "Mechanisms of quiescent nanoplastics formation in semicrystalline polymers". *Nature Communications* (**2025**), 16(1), 3051

[5] Guo G, Saidi TL, Terban MW, Valsecchi M, Billinge SJL, Lipson H. "Ab initio structure solutions from nanocrystalline powder diffraction data via diffusion models". *Nature Materials* (**2025**), 24, 1726

[6] Valsecchi M, Kumar SK. "Impact of chain ends on the amorphous phase topology in semicrystalline polymers". *Macromolecules,* (**2025**), 58(18), 10143-10163

[7] Valsecchi M, Galindo A, Jackson G. "Modelling the thermodynamic properties of the mixture of water and polyethylene glycol (PEG) with the SAFT-γ Mie group-contribution approach". *Fluid Phase Equilibria,* (**2024**), 577, 113952

[8] Valsecchi M, Galindo A, Jackson G. "Modelling sorption of hydrocarbons in polyethylene with the SAFT-γ Mie approach combined with a statistical-mechanical model to describe semi-crystalline polymers". *Macromolecules,* (**2023**), 56(24), 9952-9968

[9] Valsecchi M, Ramadani J, Williams D, Galindo A, Jackson G. "Influence of tie-molecules and microstructure on the fluid solubility in semi-crystalline polymers". *The Journal of Physical Chemistry B,* (**2022**), 126(44), 9059-9088

INVITED & CONTRIBUTED TALKS

- Invited talk @ **Institute Charles Sadron** (Strasbourg, France – 12/2024)**;**
- Contributed talks @ **AICHE** (2022, 2026), **APS** March Meeting/Global Physics Summit (2025, 2026), **ESAT** (2022, 2026) and **Thermodynamics** 2022.

PRIZES & AWARDS

- **Julia Higgins Centenary Prize –** Best PhD thesis in Chemical Engineering (Imperial, 2023).
- **Johnson Matthey Prize –** Best PhD thesis in Theory and Simulation of Materials (Imperial, 2023)
- **Outstanding GTA Student Choice Award** (Imperial College Union, 2021)
- **Best Freshmen of the A.Y. 2015/2016** (Politecnico di Milano, 2017)

MEMBERSHIPS & QUALIFICATIONS

- **Royal Society of Chemistry** – Associate Member
- **Institute of Physics** – Member